Hallucination Detection

Independent evidence verification gate, adversarial fabrication tests, measured limitations, and production safeguards

Six Anti-Hallucination Controls

Control	Mechanism
1. Independent Evidence Gate	Re-opens source PDF (PyMuPDF), BOQ (.xlsx) and Revit export (.txt/.csv). Verifies every cited snippet exists inside its exact bounding box / cell. Drops unverified anchors; rejects findings below min_evidence.
2. Minimum Evidence (≥ 2)	Cross-document conflicts and all LLM findings must cite ≥ 2 verified anchors (e.g. schedule cell + verbatim spec paragraph, or PDF schedule cell + Revit export row).
3. Zero LLM Arithmetic	All dimension chains, scale checks, tag counts (incl. TYP. OF n), electrical breaker vs MCA, and structural loads are computed in deterministic Python using exact fractions.
4. Calibrated Abstention	Ambiguous tags (bare 'TYP.') and scanned/raster pages emit 'needs_review' instead of guessing; completeness findings on raster sheets are downgraded from Confirmed to Probable.
5. Corroboration Discipline	An LLM finding is never 'Confirmed' (capped at Probable ≤ 0.85) and only corroborates a deterministic rule finding when both cite the exact same schedule cell.
6. Determinism & Audit Trail	Stable SHA-1 finding fingerprints; identical inputs produce identical outputs; every engineer Accept / Reject (with reason) is recorded in an append-only JSONL audit log.

Test 1 — Adversarial Fabricated Citations (25 Injected)

Fabrication Type (Mock LLM across 5 Spec Sections)	Injected	Blocked	Shipped
Invented spec quote	5	5	0
Non-existent element tag	5	5	0
Schedule value not in cell	5	5	0
Non-existent paragraph ID	5	5	0
Real quote with altered number	5	5	0
Total Fabrications	25	25 (100%)	0

Test 2 — Real Quote, False Claim (Measured Limitation)

Adversarial Scenario	Injected	Blocked	Outcome
Real quote + real cell + false claim	5	0	5 shipped as Probable
Wrongly merged as corroboration	5	5	0 (blocked by same-cell check)

Test 3 — LLM Client Wiring & Evaluation Harness (aecdi llm-eval)

Test Case	Measured Result
Live client vs stand-in API	Uses claude-opus-5-5, JSON schema & fallback header; refusal → abstention; truncation → error
Full pipeline (1 real + 1 fake)	Invented quote rejected by gate; valid quote corroborated rule finding on same cell
llm-eval clean mock (3 sets)	Precision 1.00, Recall 1.00, Run-to-run Consistency 1.00
llm-eval noisy mock (15% err)	Precision 0.33–0.50, Recall 0.80, Consistency 0.66 — harness catches instability

Production Monitoring & Continuous Feedback Loop

Metric / Loop	Measurement	Target	Action
Reviewer accept rate	Accepted ÷ reviewed per rule & tier	≥ 85%	< 75% → retune or disable rule
Missed-issue rate	Covered issues found later in QC / RFIs	Trend down	Root-cause & add regression case
Gate rejection rate	Rejected ÷ proposed findings	≤ 5%	> 10% → parser drift or prompt fix
Feedback → Regress	aecdi feedback → aecdi regress	0 regressions	Blocks any rule/prompt regression in CI

100% fabrication catch rate

All 25 invented quotes, wrong cell values, fake tags, fake paragraph IDs, and altered numbers were blocked by the Evidence Gate before output.

Why LLM findings are Probable

The gate proves a quote exists at the cited coordinates, not that it logically proves a conflict. Hence LLM claims are capped at Probable.

ACC governance guard

Unreviewed Probable and LLM findings are never pushed to Autodesk Construction Cloud; only engineer-accepted findings publish as Open issues.

Closed-loop regression gating

Every rejected finding becomes a fingerprinted negative test case via aecdi feedback; aecdi regress fails if a fixed false positive returns.